

Physics 513 QFT HW 2

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CONTENTS

1. 1
 - 1.1. First Attempt (Failed)
 - 1.2. Second Attempt (Succeeded)
2. 2
 - 2.1. a
 - 2.2. b
3. 3
 - 3.1. Rotations
 - 3.2. Boosts
 - 3.3. Representation
 - 3.4. Inversion
4. 4

1. 1

1.1. First Attempt (Failed)

The irreducible representation of the Lorentz group is

$$\mathbf{J}^{\pm} = \frac{\mathbf{J} \pm i\mathbf{K}}{2}, \quad (1)$$

where I use bold face for 3-vectors. Their sum is

$$\mathbf{J}^{+} + \mathbf{J}^{-} = \mathbf{J}. \quad (2)$$

This is the same as adding two angular momenta in non-relativistic quantum mechanics.

The different irreducible representations are labeled by the total angular momenta quantum numbers (j^{+}, j^{-}) . The dimensionality of a representation is

$$N = (2j^{+} + 1)(2j^{-} + 1). \quad (3)$$

The 4D representations are $(1/2, 1, 2)$, $(3/2, 0)$, and $(0, 3/2)$. Which one represents the transformations that act on 4-vectors? The

$$\left(\begin{pmatrix} 1 & 1 \\ 2 & 2 \end{pmatrix} \right) \quad (4)$$

representation. The other two represent some sort of asymmetrical pair that has nothing to do with 4-vectors which lack that kind of asymmetry. So mathematically this is just like adding two spin-1/2 angular momenta, which is discussed in Sec. 3.8 of Modern Quantum Mechanics by

Sakurai. I can use two systems of base kets. One is made from simultaneous eigenstates of $(\mathbf{J}^{\pm})^2$, and $J_z^{\pm}$

$$|j^{\pm}, m_z^{\pm}\rangle = |m^{+} m^{-}\rangle. \quad (5)$$

where I suppress the quantum number $j^{\pm}$. The four base kets in this system are

$$|++\rangle, \quad |--\rangle, \quad |+-\rangle, \quad |-+\rangle, \quad (6)$$

where $\pm$ represents $\pm 1/2$. The other system uses simultaneous eigenstates of $(\mathbf{J}^{\pm})^2$, $(\mathbf{J})^2$, and J_z

$$|j^{\pm} j m\rangle = |j m\rangle. \quad (7)$$

The 4 base kets in this system are

$$|T\rangle = |00\rangle, \quad |X\rangle = |1-1\rangle, \quad |Y\rangle = |11\rangle, \quad |Z\rangle = |10\rangle, \quad (8)$$

where T , X , Y , and Z are just labels. But I use them because I think that the distinction between spin-0 and spin-1 will correspond to the distinction between time and space in 4-vectors. It is useful to express the base kets of this system in terms of the base kets of Eq. 6. Luckily Sakurai does this for me in Eq. (3.8.15)

$$\begin{aligned} |T\rangle &= \frac{|+-\rangle - |-+\rangle}{\sqrt{2}}, \\ |X\rangle &= |--\rangle, \quad |Y\rangle = |++\rangle, \\ |Z\rangle &= \frac{|+-\rangle + |-+\rangle}{\sqrt{2}}. \end{aligned} \quad (9)$$

The operator J_z has matrix elements

$$J_z = \begin{pmatrix} (J_z)_{TT} & (J_z)_{TX} & (J_z)_{TY} & (J_z)_{TZ} \\ (J_z)_{XT} & (J_z)_{XX} & (J_z)_{XY} & (J_z)_{XZ} \\ (J_z)_{YT} & (J_z)_{YX} & (J_z)_{YY} & (J_z)_{YZ} \\ (J_z)_{ZT} & (J_z)_{ZX} & (J_z)_{ZY} & (J_z)_{ZZ} \end{pmatrix}, \quad (10)$$

where

$$(J_z)_{XT} = \langle X | J_z | T \rangle. \quad (11)$$

The matrix elements are easy to evaluate because this operator is diagonal in the Eq. 8 basis.

$$\begin{aligned} J_z |T\rangle &= J_z |00\rangle = 0, \\ J_z |X\rangle &= J_z |1-1\rangle = -|X\rangle, \\ J_z |Y\rangle &= J_z |11\rangle = |Y\rangle, \\ J_z |Z\rangle &= J_z |10\rangle = 0, \end{aligned} \quad (12)$$

$$J_z = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad (13)$$

This is very similar to the J_z I derived in HW 1. I can make the two identical by using a slightly different orthonormal basis

$$|t\rangle = |T\rangle, |z\rangle = |Z\rangle, |x\rangle = \frac{|X\rangle + |Y\rangle}{\sqrt{2}}, |y\rangle = i\frac{|X\rangle - |Y\rangle}{\sqrt{2}}. \quad (14)$$

In this basis the operator J_z has matrix elements

$$J_z = \begin{pmatrix} (J_z)_{tt} & (J_z)_{tx} & (J_z)_{ty} & (J_z)_{tz} \\ (J_z)_{xt} & (J_z)_{xx} & (J_z)_{xy} & (J_z)_{xz} \\ (J_z)_{yt} & (J_z)_{yx} & (J_z)_{yy} & (J_z)_{yz} \\ (J_z)_{zt} & (J_z)_{zx} & (J_z)_{zy} & (J_z)_{zz} \end{pmatrix} \quad (15)$$

From now on this is my convention for labeling matrix elements in all the operators. I evaluate the matrix elements.

$$J_z |t\rangle = J_z |z\rangle = 0 \quad (16)$$

$$J_z |x\rangle = \frac{J_z |X\rangle + J_z |Y\rangle}{\sqrt{2}} = \frac{-|X\rangle + |Y\rangle}{\sqrt{2}} = -\frac{|y\rangle}{i} = i|y\rangle \quad (17)$$

$$J_z |y\rangle = i\frac{J_z |X\rangle - J_z |Y\rangle}{\sqrt{2}} = i\frac{-|X\rangle - |Y\rangle}{\sqrt{2}} = -i|x\rangle \quad (18)$$

$$\langle x|J_z|x\rangle = \langle y|J_z|y\rangle = 0 \quad (19)$$

$$(J_z)_{yx} = \langle y|J_z|x\rangle = i\langle y|y\rangle = i \quad (20)$$

$$(J_z)_{xy} = \langle x|J_z|y\rangle = -i\langle x|x\rangle = -i \quad (21)$$

$$J_z = i \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad (22)$$

This is just the generator of rotations around the z axis for 4-vectors which I found in HW1. This implies that ket vectors written in the basis of Eq. 14

$$|V\rangle = V_t |t\rangle + V_x |x\rangle + V_y |y\rangle + V_z |z\rangle, \quad (23)$$

have components which transform just like the 4-vector $\vec{V} = (V_t, V_x, V_y, V_z)$. In the $(1/2, 1/2)$ representation the Lorentz real spacetime and Hilbert space transforms are the same! I rewrite the Eq. 14 basis in a more useful form.

$$\begin{aligned} |t\rangle &= \frac{|+-\rangle - |--\rangle}{\sqrt{2}} \\ |x\rangle &= \frac{|--\rangle + |++\rangle}{\sqrt{2}} \\ |y\rangle &= i\frac{|--\rangle - |++\rangle}{\sqrt{2}} \\ |z\rangle &= \frac{|+-\rangle + |--\rangle}{\sqrt{2}} \end{aligned} \quad (24)$$

I calculate the matrix elements of operators in this basis.

$$J_z^+ = \begin{pmatrix} (J_z^+)_{tt} & (J_z^+)_{tx} & (J_z^+)_{ty} & (J_z^+)_{tz} \\ (J_z^+)_{xt} & (J_z^+)_{xx} & (J_z^+)_{xy} & (J_z^+)_{xz} \\ (J_z^+)_{yt} & (J_z^+)_{yx} & (J_z^+)_{yy} & (J_z^+)_{yz} \\ (J_z^+)_{zt} & (J_z^+)_{zx} & (J_z^+)_{zy} & (J_z^+)_{zz} \end{pmatrix} \quad (25)$$

$$J_z^+ |t\rangle = \frac{J_z^+ |+-\rangle - J_z^+ |--\rangle}{\sqrt{2}} = \frac{|+-\rangle + |--\rangle}{2\sqrt{2}} = \frac{|z\rangle}{2} \quad (26)$$

$$(J_z^+)_{\mu t} = \langle \mu | J_z^+ | t \rangle = \frac{\langle \mu | z \rangle}{2} = \frac{\delta_{\mu z}}{2}, \quad \mu = t, x, y, z \quad (27)$$

The angular momentum operator is Hermitian.

$$(J_z^+)_{t\mu} = (J_z^+)_{\mu t}^\dagger = (J_z^+)_{\mu t}^* = \frac{\delta_{\mu z}}{2} \quad (28)$$

$$\begin{aligned} J_z^+ |x\rangle &= \frac{J_z^+ |--\rangle + J_z^+ |++\rangle}{\sqrt{2}} = \frac{-|--\rangle + |++\rangle}{2\sqrt{2}} \\ &= -\frac{|--\rangle - |++\rangle}{2\sqrt{2}} = -\frac{|y\rangle}{2i} = \frac{i|y\rangle}{2} \end{aligned} \quad (29)$$

$$(J_z^+)_{\mu x} = \langle \mu | J_z^+ | x \rangle = \frac{i}{2} \langle \mu | y \rangle = \frac{i}{2} \delta_{\mu y} \quad (30)$$

$$(J_z^+)_{x\mu} = (J_z^+)_{\mu x}^\dagger = (J_z^+)_{\mu x}^* = -\frac{i}{2} \delta_{\mu y} \quad (31)$$

$$\begin{aligned} J_z^+ |y\rangle &= i\frac{J_z^+ |--\rangle - J_z^+ |++\rangle}{\sqrt{2}} = -i\frac{|--\rangle + |++\rangle}{2\sqrt{2}} \\ &= -\frac{i}{2} |x\rangle \end{aligned} \quad (32)$$

$$(J_z^+)_{\mu y} = -\frac{i}{2} \langle \mu | x \rangle = -\frac{i}{2} \delta_{\mu x} \quad (33)$$

$$\begin{aligned} (J_z^+)_{zz} &= \langle z | J_z^+ | z \rangle = \langle z | \frac{J_z^+ |+-\rangle + J_z^+ |--\rangle}{\sqrt{2}} \\ &= \langle z | \frac{|+-\rangle - |--\rangle}{2\sqrt{2}} \rangle = \frac{\langle z | t \rangle}{2} = 0 \end{aligned} \quad (34)$$

$$J_z^+ = \frac{1}{2} \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & -i & 0 \\ 0 & i & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix} \quad (35)$$

$$J_z^- = \begin{pmatrix} (J_z^-)_{tt} & (J_z^-)_{tx} & (J_z^-)_{ty} & (J_z^-)_{tz} \\ (J_z^-)_{xt} & (J_z^-)_{xx} & (J_z^-)_{xy} & (J_z^-)_{xz} \\ (J_z^-)_{yt} & (J_z^-)_{yx} & (J_z^-)_{yy} & (J_z^-)_{yz} \\ (J_z^-)_{zt} & (J_z^-)_{zx} & (J_z^-)_{zy} & (J_z^-)_{zz} \end{pmatrix} \quad (36)$$

$$\begin{aligned} J_z^- |t\rangle &= \frac{J_z^- |+-\rangle - J_z^- |-+\rangle}{\sqrt{2}} = -\frac{|+-\rangle + |-+\rangle}{2\sqrt{2}} \\ &= -\frac{|z\rangle}{2} \implies (J_z^-)_{\mu t} = -\frac{\langle \mu | z \rangle}{2} = -\frac{\delta_{\mu z}}{2} \end{aligned} \quad (37)$$

$$J_z^- |x\rangle = \frac{i}{2} |y\rangle \implies (J_z^-)_{\mu x} = \frac{i}{2} \langle \mu | y \rangle = \frac{i}{2} \delta_{\mu y} \quad (38)$$

$$J_z^- |y\rangle = -\frac{i}{2} |x\rangle \implies (J_z^-)_{\mu y} = -\frac{i}{2} \langle \mu | x \rangle = -\frac{i}{2} \delta_{\mu x} \quad (39)$$

$$\begin{aligned} (J_z^-)_{zz} &= \langle z | J_z^- | z \rangle = \langle z | J_z^- \frac{|+-\rangle + |-+\rangle}{\sqrt{2}} = \\ &= \langle z | \frac{J_z^- |+-\rangle + J_z^- |-+\rangle}{\sqrt{2}} = -\langle z | \frac{|+-\rangle - |-+\rangle}{2\sqrt{2}} = \\ &= -\frac{\langle z | t \rangle}{2} = 0 \end{aligned} \quad (40)$$

$$J_z^- = \frac{1}{2} \begin{pmatrix} 0 & 0 & 0 & -1 \\ 0 & 0 & -i & 0 \\ 0 & i & 0 & 0 \\ -1 & 0 & 0 & 0 \end{pmatrix} \quad (41)$$

$$i(\mathbf{J}^- - \mathbf{J}^+) = \mathbf{K} \quad (42)$$

$$K_z = -i \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix} \quad (43)$$

This is the same z boost generator I derived in HW 1.

To get $J_x^\pm$ and $J_y^\pm$ I use the ladder operators

$$J_\pm^\pm = J_x^\pm \pm iJ_y^\pm \quad (44)$$

$$J_+^\pm = \begin{pmatrix} (J_+^\pm)_{tt} & (J_+^\pm)_{tx} & (J_+^\pm)_{ty} & (J_+^\pm)_{tz} \\ (J_+^\pm)_{xt} & (J_+^\pm)_{xx} & (J_+^\pm)_{xy} & (J_+^\pm)_{xz} \\ (J_+^\pm)_{yt} & (J_+^\pm)_{yx} & (J_+^\pm)_{yy} & (J_+^\pm)_{yz} \\ (J_+^\pm)_{zt} & (J_+^\pm)_{zx} & (J_+^\pm)_{zy} & (J_+^\pm)_{zz} \end{pmatrix} \quad (45)$$

$$J_+^\pm |t\rangle = \frac{J_+^\pm |++\rangle - J_+^\pm |--\rangle}{\sqrt{2}} \quad (46)$$

$$J_+^\pm |+-\rangle = 0 \quad (47)$$

$$J_+^\pm |t\rangle = -\frac{J_+^\pm |--\rangle}{\sqrt{2}} \quad (48)$$

$$J_+^\pm |--\rangle = |++\rangle \quad (49)$$

$$J_+^\pm |t\rangle = -\frac{|++\rangle}{\sqrt{2}} \quad (50)$$

$$\langle z | J_+^\pm | t \rangle = \langle t | J_+^\pm | t \rangle = 0 \quad (51)$$

$$\langle x | J_+^\pm | t \rangle = -\frac{\langle x | ++ \rangle}{\sqrt{2}} = -\frac{\langle -- | ++ \rangle + \langle ++ | ++ \rangle}{2} = -\frac{1}{2} \quad (52)$$

$$\langle y | J_+^\pm | t \rangle = -\frac{\langle y | ++ \rangle}{\sqrt{2}} = i\frac{\langle -- | ++ \rangle - \langle ++ | ++ \rangle}{2} = -\frac{i}{2} \quad (53)$$

$$J_+^\pm |x\rangle = \frac{|+-\rangle}{\sqrt{2}}, \quad (J_+^\pm)_{tx} = (J_+^\pm)_{zx} = \frac{1}{2} \quad (54)$$

$$J_+^\pm |y\rangle = i\frac{|+-\rangle}{\sqrt{2}}, \quad (J_+^\pm)_{ty} = (J_+^\pm)_{zy} = \frac{i}{2} \quad (55)$$

$$J_+^\pm |z\rangle = \frac{|++\rangle}{\sqrt{2}}, \quad (J_+^\pm)_{xz} = \frac{1}{2}, \quad (J_+^\pm)_{yz} = \frac{i}{2} \quad (56)$$

$$J_+^\pm = \frac{1}{2} \begin{pmatrix} 0 & 1 & i & 0 \\ -1 & 0 & 0 & 1 \\ -i & 0 & 0 & i \\ 0 & 1 & i & 0 \end{pmatrix} \quad (57)$$

$$(J_+^\pm)^\dagger = (J_x^\pm)^\dagger - i(J_y^\pm)^\dagger = J_x^\pm - iJ_y^\pm = J_-^\pm \quad (58)$$

$$J_-^\pm = \frac{1}{2} \begin{pmatrix} 0 & -1 & i & 0 \\ 1 & 0 & 0 & 1 \\ i & 0 & 0 & -i \\ 0 & 1 & -i & 0 \end{pmatrix} \quad (59)$$

$$J_-^\pm |t\rangle = \frac{|++\rangle}{\sqrt{2}}, \quad (J_-^\pm)_{xt} = \frac{1}{2}, \quad (J_-^\pm)_{yt} = \frac{i}{2} \quad (60)$$

$$J_-^\pm |x\rangle = \frac{|+-\rangle}{\sqrt{2}}, \quad (J_-^\pm)_{tx} = -\frac{1}{2}, \quad (J_-^\pm)_{zx} = \frac{1}{2} \quad (61)$$

$$J_-^\pm |y\rangle = i\frac{|+-\rangle}{\sqrt{2}}, \quad (J_-^\pm)_{ty} = -\frac{i}{2}, \quad (J_-^\pm)_{zy} = \frac{i}{2} \quad (62)$$

$$J_+^- |z\rangle = \frac{|++\rangle}{\sqrt{2}}, \quad (J_+^-)_{xz} = \frac{1}{2}, \quad (J_+^-)_{yz} = \frac{i}{2} \quad (63)$$

$$J_+^- = \frac{1}{2} \begin{pmatrix} 0 & -1 & -i & 0 \\ 1 & 0 & 0 & 1 \\ i & 0 & 0 & i \\ 0 & 1 & i & 0 \end{pmatrix} \quad (64)$$

$$J_-^+ = \frac{1}{2} \begin{pmatrix} 0 & 1 & -i & 0 \\ -1 & 0 & 0 & 1 \\ i & 0 & 0 & -i \\ 0 & 1 & -i & 0 \end{pmatrix} \quad (65)$$

$$J_+^+ + J_-^+ = J_x^+ + iJ_y^+ + J_x^+ - iJ_y^+ = 2J_x^+ \quad (66)$$

$$J_x^+ = \frac{J_+^+ + J_-^+}{2} \quad (67)$$

$$J_x^+ = \frac{1}{2} \begin{pmatrix} 0 & 0 & i & 0 \\ 0 & 0 & 0 & 1 \\ -i & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix} \quad (68)$$

$$J_x^- = \frac{1}{2} \begin{pmatrix} 0 & 0 & -i & 0 \\ 0 & 0 & 0 & 1 \\ i & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix} \quad (69)$$

$$J_+^+ - J_-^+ = J_x^+ + iJ_y^+ - J_x^+ + iJ_y^+ = 2iJ_y^+ \quad (70)$$

$$J_y^+ = i \frac{J_+^+ - J_-^+}{2} \quad (71)$$

$$J_y^+ = \frac{1}{2} \begin{pmatrix} 0 & -i & 0 & 0 \\ i & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \quad (72)$$

$$J_y^- = \frac{1}{2} \begin{pmatrix} 0 & i & 0 & 0 \\ -i & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \quad (73)$$

I derive the remaining generators.

The remaining generators did not come out right. My solution is mostly correct but I made one mistake. I chose the basis in Eq. 24 so that the z generators came out right. And I hoped that the x and y generators would work out. I should have waited until I had all the generators before doing a unitary transform that ensured all of them were correct.

1.2. Second Attempt (Succeeded)

My basis is

$$\begin{aligned} |++\rangle &= (1, 0, 0, 0), & |+-\rangle &= (0, 1, 0, 0) \\ | - + \rangle &= (0, 0, 1, 0), & |--\rangle &= (0, 0, 0, 1). \end{aligned} \quad (74)$$

I find the irreducible angular momentum operators in this basis.

$$J_z^+ = \frac{1}{2} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \quad J_z^- = \frac{1}{2} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \quad (75)$$

$$J_+^+ | - + \rangle = | ++ \rangle, \quad J_+^+ |--\rangle = | - + \rangle \quad (76)$$

$$J_+^+ = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix} \quad J_+^- = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad (77)$$

$$J_-^+ | ++ \rangle = | ++ \rangle, \quad J_-^+ | - + \rangle = | - + \rangle \quad (78)$$

$$J_-^+ = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix} \quad J_-^- = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad (79)$$

$$J_x^+ = \frac{1}{2} \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix} \quad J_x^- = \frac{1}{2} \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \quad (80)$$

$$J_y^+ = \frac{i}{2} \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix} \quad J_y^- = \frac{i}{2} \begin{pmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \end{pmatrix} \quad (81)$$

In this basis the generators are

$$\begin{aligned} J_z &= \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} & J_y &= \frac{i}{2} \begin{pmatrix} 0 & 1 & 1 & 0 \\ -1 & 0 & 0 & 1 \\ -1 & 0 & 0 & 1 \\ 0 & -1 & -1 & 0 \end{pmatrix} \\ J_x &= \frac{1}{2} \begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{pmatrix} & K_z &= i \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \\ K_y &= \frac{1}{2} \begin{pmatrix} 0 & -1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ -1 & 0 & 0 & -1 \\ 0 & -1 & 1 & 0 \end{pmatrix} & K_x &= \frac{i}{2} \begin{pmatrix} 0 & 1 & -1 & 0 \\ 1 & 0 & 0 & -1 \\ -1 & 0 & 0 & 1 \\ 0 & -1 & 1 & 0 \end{pmatrix} \end{aligned} \quad (82)$$

I need to find a unitary transform that turns these generators into the form I derived in HW 1:

2.1. a

In HW 1 I derived

$$\begin{aligned}
 \bar{J}_x &= i \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{pmatrix} & \bar{J}_y &= i \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix} \\
 \bar{J}_z &= i \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} & \bar{K}_x &= -i \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \\
 \bar{K}_y &= -i \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} & \bar{K}_z &= -i \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}.
 \end{aligned} \tag{83}$$

At this point I cheat, for the first and only time in this problem, and use the unitary transform from the solution (instead of calculating my own). [The unitary matrix from the solutions did not work. I will have to ask about this in office hours.](#)

(87)

Using these expressions I used Mathematica to confirm that

$$[P, J_i] = \{P, K_j\} = 0. \tag{88}$$

QED

2.2. b

In the $(1/2, 0)$ representation the basis states are

$$|j^+ j^- m^+ m^-\rangle = \left| \frac{1}{2} 0 m 0 \right\rangle = |m\rangle \tag{89}$$

where I suppress all quantum numbers except $m^+ = m$ because they are constants. This representation is 2D

$$x^\mu = (x^0, x^1, x^2, x^3) = (t, x, y, z) \tag{84}$$

$$(2j^+ + 1)(2j^- + 1) = (2j^+ + 1)1 = \frac{2}{2} + 1 = 1 + 1 = 2. \tag{90}$$

The basis states are

$$|+\rangle = (1, 0), \quad |-\rangle = (0, 1). \tag{91}$$

I construct the angular momentum operators for the $+$ and $-$ representations.

$$J_z^- |+\rangle = J_z^- |-\rangle = 0. \tag{92}$$

$$J_x^- = J_y^- = J_z^- = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \tag{93}$$

The $\mathbf{J}^+$ vector operator is the same as the spin-1/2 vector operator, so I can write down the answer immediately.

$$J_i^+ = \frac{\sigma_i}{2}, \tag{94}$$

In matrix notation

$$P = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}. \tag{86}$$

where the Pauli matrices are

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \sigma_y = i \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \sigma_z = i \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (95)$$

I calculate the rotation generators.

$$\mathbf{J}^\pm = \frac{\mathbf{J} \pm i\mathbf{K}}{2} \quad (96)$$

$$\mathbf{J}^+ + \mathbf{J}^- = \frac{\mathbf{J} + i\mathbf{K} + \mathbf{J} - i\mathbf{K}}{2} = \mathbf{J} \quad (97)$$

$$\mathbf{J}^+ + \cancel{\mathbf{J}^-} = \mathbf{J} \implies \mathbf{J} = \mathbf{J}^+ \quad (98)$$

$$\boxed{J_i = \frac{\sigma_i}{2}} \quad (99)$$

I calculate the boost generators.

$$\mathbf{J}^+ - \cancel{\mathbf{J}^-} = i\mathbf{K} \implies i\mathbf{K} = \mathbf{J}^+ = \mathbf{J} \implies \mathbf{K} = -i\mathbf{J} \quad (100)$$

$$\boxed{K_j = -\frac{i}{2}\sigma_j} \quad (101)$$

I assume that the parity matrix can exist. This matrix must satisfy that $[P, J_i] = 0$.

$$[P, J_i] = \left[P, \frac{\sigma_i}{2} \right] = \frac{[P, \sigma_i]}{2} \quad (102)$$

$$[P, J_i] = 0 \implies [P, \sigma_i] = 0 \quad (103)$$

It must also satisfy $\{P, K_i\} = 0$.

$$\{P, K_i\} = \left\{ P, -\frac{i}{2}\sigma_i \right\} = -\frac{i}{2}\{P, \sigma_i\} \quad (104)$$

$$\{P, K_i\} = 0 \implies \{P, \sigma_i\} = 0. \quad (105)$$

No nonzero matrix can satisfy

$$\{P, \sigma_i\} = [P, \sigma_i] = 0. \quad (106)$$

Therefore the parity operation does not exist in the $(1/2, 0)$ representation. QED.

3. 3

3.1. Rotations

I define

$$\mathbf{G}_\pm = \mathbf{E} \pm i\mathbf{B}. \quad (107)$$

Because the electric, $\mathbf{E}$ and magnetic, $\mathbf{B}$, fields are 3-vectors and the $\mathbf{G}_\pm$ are linear combinations of $\mathbf{E}$ and $\mathbf{B}$ they are also 3-vectors. Since 3-vectors are invariant under rotations $\mathbf{G}_\pm$ are invariant under rotations. QED.

3.2. Boosts

The electric and magnetic fields cannot be described by 4-vectors. To study their transformations under boosts I need the electromagnetic field tensor F . My coordinates are

$$x^\mu = (x^0, x^1, x^2, x^3) = (t, x, y, z). \quad (108)$$

The metric is

$$\eta_{\mu\nu} = \text{diag}(1, -1, -1, -1). \quad (109)$$

The electromagnetic field in these coordinates is

$$F^{\mu\nu} = \begin{pmatrix} 0 & -E_x & -E_y & -E_z \\ E_x & 0 & -B_z & B_y \\ E_y & B_z & 0 & -B_x \\ E_z & -B_y & B_x & 0 \end{pmatrix}. \quad (110)$$

This corresponds to 3-vectors

$$\mathbf{E} = (E_x, E_y, E_z), \quad \mathbf{B} = (B_x, B_y, B_z). \quad (111)$$

I will only consider x boost. Because space is isomorphic in SR I lose no generality. My boosted coordinates are

$$x^{\bar{\mu}} = (x^{\bar{0}}, x^{\bar{1}}, x^{\bar{2}}, x^{\bar{3}}) = (\bar{t}, \bar{x}, \bar{y}, \bar{z}). \quad (112)$$

The Lorentz boost transformation is

$$\Lambda_{\mu}^{\bar{\mu}} = \begin{pmatrix} \gamma & -\gamma v & 0 & 0 \\ -\gamma v & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad (113)$$

where v is the boost velocity and

$$\gamma = \frac{1}{\sqrt{1-v^2}}. \quad (114)$$

I find the electric and magnetic 3-vectors in the boosted frame.

$$F^{\bar{\mu}\bar{\nu}} = \begin{pmatrix} 0 & -E_{\bar{x}} & -E_{\bar{y}} & -E_{\bar{z}} \\ E_{\bar{x}} & 0 & -B_{\bar{z}} & B_{\bar{y}} \\ E_{\bar{y}} & B_{\bar{z}} & 0 & -B_{\bar{x}} \\ E_{\bar{z}} & -B_{\bar{y}} & B_{\bar{x}} & 0 \end{pmatrix} \quad (115)$$

$$\bar{\mathbf{E}} = (E_{\bar{x}}, E_{\bar{y}}, E_{\bar{z}}), \quad \bar{\mathbf{B}} = (B_{\bar{x}}, B_{\bar{y}}, B_{\bar{z}}) \quad (116)$$

$$F^{\bar{\mu}\bar{\nu}} = \Lambda_{\mu}^{\bar{\mu}} \Lambda_{\nu}^{\bar{\nu}} F^{\mu\nu} \quad (117)$$

$$E_{\bar{i}} = F^{\bar{i}\bar{0}} = \Lambda_{\mu}^{\bar{i}} \Lambda_{\nu}^{\bar{0}} F^{\mu\nu} = \Lambda_{\mu}^{\bar{i}} \Lambda_{\nu}^{\bar{0}} F^{\mu 0} + \Lambda_{\mu}^{\bar{i}} \Lambda_{\nu}^{\bar{0}} F^{\mu 1} = \gamma \Lambda_{\mu}^{\bar{i}} F^{\mu 0} - \gamma v \Lambda_{\mu}^{\bar{i}} F^{\mu 1} \quad (118)$$

$$E_{\bar{i}} = \gamma(\Lambda_{\mu}^{\bar{i}} F^{\mu 0} - v \Lambda_{\mu}^{\bar{i}} F^{\mu 1}) \quad (119)$$

$$\begin{aligned}
E_{\bar{x}} &= \gamma(\cancel{\Lambda_{\bar{x}}^{\bar{x}} F^{tt}} + \Lambda_{\bar{x}}^{\bar{x}} F^{xt} - v\Lambda_{\bar{x}}^{\bar{x}} F^{tx} - \cancel{v\Lambda_{\bar{x}}^{\bar{x}} F^{xx}}) \\
&= -\gamma(\Lambda_{\bar{x}}^{\bar{x}} + v\Lambda_{\bar{x}}^{\bar{x}} F^{tx}) = \gamma^2(1-v^2)E_x = \frac{\gamma^2}{\gamma^2}E_x = E_x
\end{aligned} \tag{120}$$

$$\begin{aligned}
E_{\bar{y}} &= \gamma(\Lambda_{\bar{\mu}}^{\bar{y}} F^{\mu t} - v\Lambda_{\bar{\mu}}^{\bar{y}} F^{\mu x}) = \\
&\quad \gamma(F^{yt} - vF^{yx}) = \gamma(E_y - vB_z)
\end{aligned} \tag{121}$$

$$E_{\bar{z}} = \gamma(F^{zt} - vF^{zx}) = \gamma(E_z + vB_y) \tag{122}$$

$$B_{\bar{x}} = F^{\bar{z}\bar{y}} = \Lambda_{\bar{\mu}}^{\bar{z}} \Lambda_{\bar{\nu}}^{\bar{y}} F^{\mu\nu} = \Lambda_{\bar{z}}^{\bar{z}} \Lambda_{\bar{y}}^{\bar{y}} F^{zy} = F^{zy} = B_x \tag{123}$$

$$\begin{aligned}
B_{\bar{y}} &= F^{\bar{x}\bar{z}} = \Lambda_{\bar{\mu}}^{\bar{x}} \Lambda_{\bar{\nu}}^{\bar{z}} F^{\mu\nu} = \Lambda_{\bar{\mu}}^{\bar{x}} F^{\mu z} = \Lambda_{\bar{t}}^{\bar{x}} F^{tz} + \Lambda_{\bar{x}}^{\bar{x}} F^{xz} \\
&= \gamma(-vF^{tz} + F^{xz}) = \gamma(vE_z + B_y)
\end{aligned} \tag{124}$$

$$B_{\bar{z}} = F^{\bar{y}\bar{x}} = \gamma(B_z - vE_y) \tag{125}$$

$$\begin{aligned}
&\boxed{E_{\bar{x}} = E_x, \quad E_{\bar{y}} = \gamma(E_y - vB_z), \quad E_{\bar{z}} = \gamma(E_z + vB_y)} \\
&\boxed{B_{\bar{x}} = B_x, \quad B_{\bar{y}} = \gamma(vE_z + B_y), \quad B_{\bar{z}} = \gamma(B_z - vE_y)}
\end{aligned} \tag{126}$$

I will show that the components of $\mathbf{G}_{\pm}$ transform among themselves under boosts. In the original frame

$$\mathbf{G}_{\pm} = (G_{\pm}^x, G_{\pm}^y, G_{\pm}^z) = (E_x \pm iB_x, E_y \pm iB_y, E_z \pm iB_z). \tag{127}$$

In the boosted frame

$$\bar{\mathbf{G}}_{\pm} = (G_{\pm}^{\bar{x}}, G_{\pm}^{\bar{y}}, G_{\pm}^{\bar{z}}) = (E_{\bar{x}} \pm iB_{\bar{x}}, E_{\bar{y}} \pm iB_{\bar{y}}, E_{\bar{z}} \pm iB_{\bar{z}}). \tag{128}$$

I find how the two sets are related.

$$G_{\pm}^{\bar{x}} = E_{\bar{x}} \pm iB_{\bar{x}} = E_x \pm iB_x = G_{\pm}^x \tag{129}$$

$$\boxed{G_{\pm}^{\bar{x}} = G_{\pm}^x} \tag{130}$$

$$\begin{aligned}
G_{\pm}^{\bar{y}} &= E_{\bar{y}} \pm iB_{\bar{y}} = \gamma(E_y - vB_z) \pm i\gamma(vE_z + B_y) = \\
&\gamma(E_y - vB_z \pm viE_z \pm iB_y) = \gamma(E_y \pm iB_y - vB_z \pm viE_z) = \\
&\gamma[G_{\pm}^y \pm v(iE_z \mp B_z)] = \gamma[G_{\pm}^y \pm iv(E_z \pm iB_z)] = \\
&\quad \gamma(G_{\pm}^y \pm ivG_{\pm}^z)
\end{aligned} \tag{131}$$

$$\boxed{G_{\pm}^{\bar{y}} = \gamma(G_{\pm}^y \pm ivG_{\pm}^z)} \tag{132}$$

$$\begin{aligned}
G_{\pm}^{\bar{z}} &= E_{\bar{z}} \pm iB_{\bar{z}} = \gamma(E_z + vB_y) \pm i\gamma(B_z - vE_y) = \\
&\gamma(E_z + vB_y \pm iB_z \mp ivE_y) = \gamma(E_z \pm iB_z + vB_y \mp ivE_y) = \\
&\gamma[G_{\pm}^z + v(B_y \mp iE_y)] = \gamma[G_{\pm}^z + v(\mp iE_y + B_y)] \\
&= \gamma[G_{\pm}^z \mp iv(E_y \pm iB_y)] = \gamma(G_{\pm}^z \mp ivG_{\pm}^y)
\end{aligned} \tag{133}$$

$$\boxed{G_{\pm}^{\bar{z}} = \gamma(G_{\pm}^z \mp ivG_{\pm}^y)} \tag{134}$$

Thus the two sets transform among themselves. QED.

3.3. Representation

Because the $\vec{G}_{\pm}$ are 4-vectors they transform under the (1/2,1/2) representation (which I showed in Problem 1).

3.4. Inversion

The inversion transform is

$$P_{\mu}^{\bar{\mu}} = \text{diag}(1, -1, -1, -1). \tag{135}$$

It has the same components as the metric. The inverted frame is

$$x^{\mu'} = (x^{0'}, x^{1'}, x^{3'}, x^{4'}) = (t', x', y', z'). \tag{136}$$

I will find the electric and magnetic fields in this frame.

$$F^{\mu'\nu'} = P_{\mu}^{\mu'} P_{\nu}^{\nu'} F^{\mu\nu} \tag{137}$$

$$E_{i'} = F^{i't'} = P_{\mu}^{i'} P_{\nu}^{t'} F^{\mu\nu} = P_i^{i'} P_t^{t'} F^{it} = -F^{it} = -E_i \tag{138}$$

Thus under parity

$$\boxed{\mathbf{PE} = -\mathbf{E}}. \tag{139}$$

$$\begin{aligned}
B_{x'} &= F^{z'y'} = P_{\mu}^{z'} P_{\nu}^{y'} F^{\mu\nu} = P_z^{z'} P_y^{y'} F^{zy} = \\
&\quad \cancel{(-1)^2} F^{zy} = B_x
\end{aligned} \tag{140}$$

$$B_{y'} = F^{x'z'} = P_x^{x'} P_z^{z'} F^{xz} = F^{xz} = B_y \tag{141}$$

$$B_{z'} = F^{y'x'} = B_z \tag{142}$$

$$\boxed{\mathbf{PB} = \mathbf{B}}. \tag{143}$$

$$P\mathbf{G}_{\pm} = \mathbf{PE} \pm i\mathbf{PB} = -\mathbf{E} \pm i\mathbf{B} = -(\mathbf{E} \mp i\mathbf{B}) = -\mathbf{G}_{\mp} \tag{144}$$

$$\boxed{P\mathbf{G}_{\pm} = -\mathbf{G}_{\mp}} \tag{145}$$

4. 4

$$L_1 = (\vec{a}_1, \Lambda_1), \quad L_2 = (\vec{a}_2, \Lambda_2) \quad (146)$$

$$L_3 = L_2 L_1 = (\vec{a}_2 + \Lambda_2 \vec{a}_1, \Lambda_2 \Lambda_1) \quad (147)$$

$$L_2 L_1 \vec{x} = L_3 \vec{x} \quad (148)$$

$$L_1 \vec{x} = \Lambda_1 \vec{x} + \vec{a}_1 \quad (149)$$

$$L_2(\Lambda_1 \vec{x} + \vec{a}_1) = (\vec{a}_2 + \Lambda_2 \vec{a}_1, \Lambda_2 \Lambda_1) \vec{x} \quad (150)$$

$$L_2(\Lambda_1 \vec{x} + \vec{a}_1) = \Lambda_2 \Lambda_1 \vec{x} + \vec{a}_2 + \Lambda_2 \vec{a}_1 \quad (151)$$

The Poincaré transformation is not linear because

$$L_2(\Lambda_1 \vec{x} + \vec{a}_1) \neq L_2 \Lambda_1 \vec{x} + L_2 \vec{a}_1, \quad (152)$$

leads to the wrong answer. Instead the correct move is

$$\Lambda_2(\Lambda_1 \vec{x} + \vec{a}_1) + \vec{a}_2 = \Lambda_2 \Lambda_1 \vec{x} + \vec{a}_2 + \Lambda_2 \vec{a}_1. \quad (153)$$

$$\Lambda_2(\Lambda_1 \vec{x} + \vec{a}_1) + \cancel{\vec{a}_2} = \Lambda_2 \Lambda_1 \vec{x} + \cancel{\vec{a}_2} + \Lambda_2 \vec{a}_1 \quad (154)$$

$$\cancel{\Lambda_2 \Lambda_1 \vec{x} + \Lambda_2 \vec{a}_1} = \cancel{\Lambda_2 \Lambda_1 \vec{x} + \Lambda_2 \vec{a}_1} \quad (155)$$

$$0 = 0 \quad (156)$$

QED